# PROCESS DESIGN DOCUMENT (PDD)

## Returns Automation Agent

### Glendale Parade Store LLC

Prepared by Beecker | Version 1.0 | February 24, 2026

### Document Purpose

**This document describes how the Glendale Returns Automation Agent ('Agent') is expected to behave once in production. It is written to be reviewed and approved by Glendale Parade Store stakeholders.**

- The goal is to clearly define:
- What the Agent monitors and triggers on
- What is automated versus what requires human intervention
- What communications the Agent sends to customers
- What actions the Agent performs in connected systems (MOM ERP, UPS Web API)
- When the Agent is considered to be working correctly (success criteria)

This document will serve as the **source of truth** for the Agent's behavior once approved. Items marked [TBD] require confirmation from the relevant stakeholder before development begins.

## 1. Scope & Objectives

### Objective

The Returns Automation Agent is designed to streamline and automate the end-to-end returns process for Glendale Parade Store. Currently, the returns workflow is highly inefficient: information is missing in 90% of cases, requiring customer service to spend significant time chasing customers via phone and email. The Agent eliminates this friction by:

- Centralizing all return requests through a single web form on glendale.com
- Automating return execution for requests below a defined dollar threshold
- Sending professional, non-robotic customer communications throughout the process
- Triggering the appropriate actions in MOM (ERP/Inventory) and UPS (shipping labels via web API)

### The Agent Does NOT

- Override approvals – once a return is approved, it is final and the script executes
- Make inventory decisions – MOM manages all inventory changes
- Handle RMA/Exchange automation – this process remains manual due to complexity

## 2. High-Level Agent Behavior

At a high level, the Agent Workflow performs the following steps:

1. Customer submits a return request via the web form at glendale.com (or Customer Service fills it out on the customer's behalf during a phone call).
2. The form submission populates a dedicated **MySQL returns table** in Glendale's backend.
3. The Glendale team checks whether all required information on each form submission is present. If any required field is missing the Glendale team marks the return as 'Needs Attention.
4. The Glendale team checks all open 'Needs Attention' requests to:
    a. Approve a return
    b. Add comments on a return (reason why it cannot be approved)
    c. Reject a return
5. The Agent reads from this returns table once at the end of the day.
6. The Agent checks all requests placed as 'Pending - Complete Info',
    a. For returns **below the auto-approval threshold** [TBD – $150], the Agent auto-approves and moves them into the 'Approved' tab.
    b. For returns above the threshold, the Agent moves them into the 'Needs Attention' tab for human review.
7. The Agent checks all 'Needs Attention' requests
    a. If the request has comments added by Glendale team, the agent sends an email communication to the customer informing them the request cannot be processed because of the reason specified in the comments. The status remains as 'Needs Attention'.
        i. The email is extracted from the same row belonging to the return request.
        ii. The message is constructed using a LLM and the comments as context info.
        iii. Email must be sent at normal working hours, not late night.
    b. If the request has NO comments, the agent notifies the Glendale team that there is a pending request in the 'Needs Attention' tab that needs to be either approved or revised (add comments).The status remains as 'Needs Attention'.
8. For all the 'Approved' requests, the Agent executes the appropriate script based on the return type (see Section 4). MOM actions are triggered either via nightly CSV ingestion (Option A) or real-time API (Option B) – pending confirmation from Jill.
9. The Agent executes the appropriate script based on the return type (see Section 4). MOM actions are triggered either via nightly CSV ingestion (Option A) or real-time API (Option B)
10. The Agent sends the appropriate customer communication at each stage (submission confirmation, approval/rejection, shipping label where applicable).

11. If the Agent encounters a scenario it cannot handle (e.g., customer replies with complex queries), it raises an alert for human intervention.
    a. Flag raised in run history
    b. Notification summary to Glendale team at the end of agent execution

# 3. Systems Involved

| System | Role | Integration Method |
|---|---|---|
| Glendale Website (glendale.com) | Return form intake & backend admin | Read-only DB access (MySQL credentials) – provided by Vishal |
| MySQL Returns Table | Source of truth for all return requests | Direct DB read via SQL Workbench / Agent connector |
| MOM ERP | Inventory management, order processing, RMA generation | Two options under review:<br>• Option A: Nightly CSV file ingestion via MOM API – Agent queues actions end-of-day; MOM processes overnight; emails sent next morning.<br>• Option B: Direct API / database access – Agent triggers MOM actions in real time, enabling same-day execution.[FLAG: to confirm which is technically feasible and preferred.] |
| UPS Web API | Generate and deliver return shipping labels (replaces WorldShip desktop) | UPS web-based API / web service [Pending UPS confirmation of API access & pricing] |
| Email System (Outlook) | Customer communications (confirmations, approvals, labels) | Agent-managed; non-robotic tone required. “noreply@...” style email must be provided with credentials to it |

# 4. Feature Breakdown

## Feature 1: Daily Return Table Intake & Status Segmentation

### What the Agent Does

The Agent reads from Glendale’s MySQL **returns table once per business day at the end of day**. All return requests, regardless of origin (web form or Customer Service phone call), are already normalized into this table via the web form at glendale.com.

The Agent does **not** validate completeness. That responsibility belongs to the Glendale team earlier in the process.

The Agent focuses only on requests that have already been classified by Glendale as:

- **Pending – Complete Info**
- **Needs Attention**

### Trigger

Once per business day (Monday through Friday), at the end of day.

### Logic

- Agent queries the returns table
  - Agent performs a single read of the MySQL returns table at the end of business day.
  - Agent retrieves all return records whose status is either:
    - Pending – Complete Info
    - Needs Attention
  - All other statuses are ignored.
- Agent processes returns with status “Pending – Complete Info”
  - For each return marked “Pending – Complete Info”, the Agent reads the return dollar value.
  - Agent compares the return value against the configured auto-approval threshold ($150)
    - **If the return value is below the threshold:** Agent updates the return status to Approved.
    - **If the return value is equal to or above the threshold**: Agent updates the return status to “Needs Attention” to route the request back to the Glendale team for human review.
  - Threshold applies only to:
    - Damaged Goods
    - Wrong Item
    - Manufacturer Defect
- Agent processes returns with status “Needs Attention”
  - For each return marked “Needs Attention”, the Agent checks whether Glendale team comments are present on the return record.
    - If comments are present:
      - Agent extracts the customer email address from the same return record.
      - Agent generates a customer-facing email using an LLM, using the Glendale team comments as contextual input.
      - Agent sends the email to the customer during normal working hours.
      - Return status remains “Needs Attention”.
    - If comments are not present:

            - Agent sends an internal notification to the Glendale team indicating that a return in “Needs Attention” requires action (approve, reject, or add comments).
            - No customer communication is sent.
            - Return status remains “Needs Attention”.
- Agent processes returns with status “Rejected”
    - For each return marked “Rejected”, the Agent checks whether Glendale team comments are present on the return record.
        - If comments are present:
            - Agent extracts the customer email address from the same return record.
            - Agent generates a customer-facing email using an LLM, using the Glendale team comments as contextual input.
            - Agent sends the email to the customer during normal working hours.
            - Return is archived/removed from “Rejected”.
        - If comments are not present:
            - Agent sends an internal notification to the Glendale team indicating that a return in “Rejected” requires action (add comments).
            - No customer communication is sent.
            - Return status remains “Needs Attention”.
- Status integrity
    - The Agent does not modify or override Glendale team decisions.
    - The Agent does not change the status of “Needs Attention” returns outside of the threshold-routing logic.
    - Each return is processed once per daily Agent run.

**Success Criteria**

- Agent successfully ingests all new and updated return requests from the MySQL table each business day.
- No return request with status **Pending – Complete Info** or **Needs Attention** is skipped.
- No duplicate processing of the same request within a single day.
- 100% of below-threshold returns are auto-approved without human intervention.
- 100% of above-threshold returns are routed to Needs Attention within the same business day.
- No returns bypass the threshold logic.

## Feature 2: Automated Customer Communications

**What the Agent Does**

The Agent is responsible for sending customer-facing emails at each stage of the returns process. Emails must be professional, empathetic, and non-robotic in tone. If a customer replies with a complex query, the Agent escalates to a human team member.

**Email Events**

| Trigger Event | Email Content | Tone |
|---|---|---|
| Needs Attention (missing info) | We noticed some information is missing from your return request. Please provide [X] so we can move forward. | Helpful, non-accusatory |
| Approved | Great news – your return request has been approved. [Next steps based on return type]. | Positive, clear next steps |
| Shipping label issued (where applicable) | Please find your prepaid return shipping label attached. [Instructions]. | Clear, action-oriented |
| Rejected | After review, we were unable to approve your return request for the following reason: [reason]. Please contact us if you have questions. | Respectful, with recourse option |

## Feature 3: Damaged Goods & Manufacturer Defect Processing

**Context**

Damaged Goods and Manufacturer Defect follow an identical process. Once approved (manually or auto-approved by the threshold), the Agent executes the following script.

**Execution Steps**

1. Generate a return shipping label via the UPS Web API (replaces manual WorldShip process). The label is saved as a PDF.
2. Email the shipping label PDF to the customer, so they can ship the damaged/defective item back to Glendale.
3. Execute the corresponding MOM actions via back-end ERP integration:
    a. Navigate to the order, select the affected line item and quantity, set 'Item is an Exchange,' and select reason 'Damage Not Usable.' This prevents the damaged item from re-entering active inventory and flags it for supplier return review.

**MOM Reason Code (TBD)**

| Return Type | MOM Reason Code |
|---|---|
| Damaged Goods | Damage Not Usable |
| Manufacturer Defect | Damage Not Usable |

**Flagged If**

- UPS API call fails – Agent logs error and alerts human team to generate label manually
- MOM action fails – Agent logs error and alerts human team

**Success Criteria**

- Shipping label is generated via UPS API and emailed to customer
- MOM is updated with correct reason code and item is marked committed for reshipment

## Feature 4: Wrong Item Processing

**Context**

The Wrong Item process is nearly identical to Damaged Goods, with one key difference in the MOM reason code. The incorrect item is returned to inventory upon receipt (rather than flagged as damaged) so it can be redistributed.

**Execution Steps**

1. Generate return shipping labels via UPS Web API and save as PDF.
1. Email the shipping label PDF to the customer.
2. Execute the corresponding MOM actions via back-end ERP integration:
   Navigate to the order, select the affected line item and quantity, set 'Item is an Exchange,' and select reason 'Wrong Item Shipped.' This returns the item to active inventory and immediately triggers reshipment of the correct item.
3. Print the pick ticket in MOM to trigger reshipment.

**MOM Reason Code (TBD)**

| Return Type | MOM Reason Code |
|---|---|
| Wrong Item | Wrong Item Shipped |

**Key Difference from Damaged Goods**

The 'Wrong Item Shipped' reason code in MOM places the returned item back into active inventory, enabling immediate reshipment. 'Damage Not Usable' does not return the item to active inventory.

## Feature 5: Refund Processing

### Context

A Refund applies when a customer received the correct item but no longer wants it (e.g., changed mind). Unlike Damaged or Wrong Item returns, Glendale does NOT provide a shipping label for refunds. The customer is responsible for shipping the item back at their own cost via UPS.

### Execution Steps

- **Step 1 – Trigger:** Refund returns require manual approval by Glendale's team. ***No auto-approval by threshold applies to refunds***, as the item must be physically reviewed upon arrival. The agent will be triggered on the "Pending - Complete" status, only once the user has physically verified will they change status to "Approved" manually.
- **Step 2 – Customer Notification:** Agent sends an email instructing the customer to ship the item back to Glendale at their own expense. No shipping label is provided.
- **Step 3 – Item Receipt & Review:** Warehouse team (Julian) physically reviews the returned item upon arrival to confirm it is serviceable. This is a manual step and cannot be automated.
- **Step 4a – Item passes review:** Once the item passes review, the approver sets status to 'Approved'
- **Step 4b – Item does NOT pass review:** The approver sets status to 'Rejected' and adds corresponding comments
- **Step 5 – MOM Action:** Agent executes the MOM RMA process: navigate to Order Edits > Return Merchandise Authorization, enter return quantity, select reason 'Changed Their Mind,' add notes, save to generate the RMA number, then process to issue the refund and return the item to inventory.

### Note – No Shipping Label

The UPS Web API is NOT called for Refund returns. The customer bears all return shipping costs.

### Success Criteria

- Customer is notified of approval and shipping instructions without a prepaid label
- Item is reviewed manually upon receipt before refund is processed
- Refund is issued in MOM and the item is returned to inventory

## Feature 6: Missing Item Processing

### Context

A Missing Item claim occurs when a customer reports that fewer items than ordered were received. This process includes a required manual warehouse verification step before any action is taken.

### Execution Steps

- **Step 1 – Trigger:** Customer (or CS) submits a return form specifying the missing item(s), quantity, and invoice number. Missing item requests require manual approval by the warehouse team (Julian). ***No auto-approval by threshold applies to missing items***, as the discrepancy must be physically verified against inventory.
- **Step 2 – Warehouse Count:** Warehouse team (Julian) physically goes to the warehouse, locates the bin for the relevant SKU, and counts available inventory to verify the claim.
- **Step 3a – Count Confirms Missing:** If the internal count confirms the discrepancy, warehouse team must mark the missing item request as “Approved” the Agent proceeds to Step 4.
- **Step 3b – Count Does Not Confirm:** If inventory count is correct, the warehouse team must mark the missing item request as “Rejected” and add the reason in comments. The Agent sends a follow-up email to the customer asking them to re-verify (e.g., 'Could you double-check your delivery? Sometimes items are packed in separate boxes.'). Glendale policy is to never deny the customer outright.
- **Step 4 – MOM Action:** Agent executes the MOM action via back-end integration (NOT UI): navigate to the order, select the missing line item, set 'Item is an Exchange,' and select reason 'Missing Item.' This places the item back into inventory for immediate reshipment.

### Note – No Shipping Label

The UPS Web API is NOT called for Missing Item returns. The replacement item ships as a new outbound order from Glendale's warehouse.

## 5. Return Status Workflow

All returns flow through the following status values, managed via the 'Status' column in the Glendale MySQL backend:

| Status | Set By | Meaning |
|---|---|---|
| Pending | System (on form submission) | Return submitted; awaiting information check or manual review |
| Needs Attention | Agent (above threshold) or Approver (missing info + comments) | Required information is missing; follow-up email sent to customer or approval request sent to Glendale team |
| Approved | Agent (auto) or Approver (manual) | Return approved; Agent will execute the return script |
| Rejected | Warehouse or Missing Item/Refund Approver | A missing item or refund request has been rejected because of business rules; Agent will send communication to customer |

**IMPORTANT: Once a return is set to 'Approved' and the script begins execution, the action is final. There is no reversal once the script runs.**

## 6. Out of Scope

The following are explicitly out of scope for this phase of the project:

- RMA / Exchange Automation – requires too many manual human decisions (SKU identification, pricing negotiation, serviceability review)
- WorldShip Desktop Application – replaced entirely by UPS Web API; no UI automation on the WorldShip desktop app
- UI Integration with MOM ERP – replaced by backend integration
- Returns from channels other than the web form – all returns must go through the web form; historical returns not submitted via the form are out of scope

# 7. Open Items & Assumptions

The following items require confirmation before development begins. They are flagged throughout this document.

| # | Item | Details | Owner | Target Date |
|---|---|---|---|---|
| 1 | Auto-Approval Threshold | CFO negotiation in progress. Range: $150–$200. Final value needed before development. | Jose Barreto / CFO | TBD |
| 2 | MOM Integration Method | Two options to evaluate: (MOM expert):<br>● Option A – Nightly CSV: Agent generates a CSV of approved actions end-of-day; MOM ingests overnight; customer emails queue and send next morning. Lower technical risk.<br>● Option B – Direct API/DB: Agent calls MOM API in real time for immediate execution and same-day customer communications. Higher value but more complex.<br>Decision needed before development sprint begins. | Jose Barreto / Team | ASAP |
| 3 | UPS Web API Access | Jose Barreto meeting with UPS rep to confirm web-based access and API/web service availability for programmatic label generation. Cost of API calls also to be confirmed. | Jose Barreto / UPS Rep | ASAP |
| 5 | Email Template Content | Specific email copy for each event (submission, approval, rejection, shipping | Jose Barreto / Jonathan Piza | TBD |

| # | Item | Details | Owner | Target Date |
|---|---|---|---|---|
| | | label, etc.) to be written and approved by Glendale. | | |

## 8. Approval & Sign-Off

By approving this document, stakeholders confirm that:

- The described processes accurately reflect Glendale's current and intended future returns workflow
- The automation scope and exclusions are acceptable for this phase
- Open items will be resolved by the dates noted in Section 7
- Items outside this scope will be treated as change requests

| Name | Title | Date | Signature |
|---|---|---|---|
| Jose Barreto | CEO – Glendale Parade Store LLC | | |
| Jonathan Piza | Product Manager – Beecker | | |
| Antonio Marbán | Product Manager – Beecker | | |